TECHNICAL REPORT TR-2026-05

Silent-Failure Study

Three Problem Families, and One Engine-Specific Defect

Noah Parsons

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Engines: MechanicsDSL 2.1.3 (`a8dc2b2`), `sympy.physics.mechanics` 1.14.0, Drake 1.56.0.

Referee: closed-form `numpy` references, sharing no library with any engine under test.

Disclosure: the author maintains one of the engines measured. AI assistance was used. See §7.

Abstract

The study's principal weakness was that every result came from one planar pendulum chain. Two structurally different families were therefore added: a slider–crank, which introduces a prismatic joint, a closed loop, and dead centres where the effective inertia collapses for geometric reasons; and a cart–pole, a tree with mixed joint types and a configuration-dependent mass matrix whose determinant collapses when the cart is light. Across 72 engine–case pairs spanning both new families there were no disagreements and no refusals: all three engines matched the independent references to 1.3×10^{-14} or better, on derivation and on integration, including at the degenerate configurations. One genuine engine-specific defect was found and is reported in §5: Drake's continuous-mode API accepts a distance constraint and silently omits it from the dynamics. Its provenance is traced to a specific commit, its version range is bounded by tags, and the maintainers' own stated intent is recorded, which narrows the finding considerably. §6 documents a diagnostic that caught three of this study's own errors and is the most transferable result here.

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1 Why more families

Everything through Week 1 rested on one planar pendulum chain plus two constrained systems built from the same ingredients: point masses, revolute freedom, quadratic constraints. A reviewer would reasonably ask whether the agreement observed between engines is a property of dynamics engines or a property of that problem, and on that evidence the question could not be answered.

Two families were added, chosen so that each exercises machinery the chain does not and each degenerates by a different mechanism.

Family	New machinery	Degeneracy
Pendulum chain (1)	—	none intrinsic; amplitude drives it into chaos
Slider–crank (2)	prismatic joint; closed loop	dead centre: transmission ratio $\mathrm{d}x/\mathrm{d}\theta \to 0$
Cart–pole (3)	mixed joints in a tree; coupled configuration-dependent mass matrix	light cart: $\det M \to 0$ at vertical

2 Family 2: the slider–crank

2.1 Construction

Crank pivot at the origin, crank length r, angle θ; crank pin at $P = (r\cos\theta, r\sin\theta)$; slider at $S = (x, 0)$ on the x-axis; massless rod of length l joining them. Coordinates $q = (\theta, x)$, one constraint, one degree of freedom. The constraint collapses pleasantly:

$$g = |S - P|^2 - l^2 = x^2 - 2rx\cos\theta + r^2 - l^2, \tag{1}$$

the $\sin^2 + \cos^2$ becoming a constant, so J and $\dot{J}\dot{q}$ have closed forms and the reference needs no symbolic algebra.

2.2 Why dead centre is the point

At $\theta = 0$ and $\theta = \pi$ the crank and rod are collinear and $\mathrm{d}x/\mathrm{d}\theta = 0$: the slider is instantaneously stationary however fast the crank turns, so its mass contributes nothing to the effective inertia

$$M_{\text{eff}}(\theta) = m_c r^2 + m_s \left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2. \tag{2}$$

With a heavy slider this collapses twice per revolution — by 1.1×10^2 at $m_s/m_c = 10^2$ and 1.1×10^6 at 10^6. Unlike the suite's `near_singular` axis, which degrades a hand-built mass matrix, this degeneracy is produced by geometry in normal operation.

2.3 Results

Two knobs: rod-to-crank ratio l/r from 3.0 toward the folding configuration at 1.01, and slider-to-crank mass ratio to 10^6. Each engine checked at 16 probe states on the manifold *and* evaluated exactly at both dead centres, then integrated for 5 s under a pinned `DOP853`.

l/r	m_s/m_c	Engine	EOM vs. ref	Energy drift	Constraint
3	10^2	MechanicsDSL	1.22×10^{-15}	1.51×10^{-10}	1.81×10^{-10}
3	10^2	SymPy	5.55×10^{-16}	1.51×10^{-10}	1.81×10^{-10}
3	10^2	Drake (disc.)	—	—	1.77×10^{-6}
1.5	10^2	MechanicsDSL	4.17×10^{-15}	2.11×10^{-10}	9.12×10^{-10}
1.05	10^2	MechanicsDSL	5.66×10^{-15}	5.40×10^{-10}	1.40×10^{-9}
3	10^4	MechanicsDSL	1.97×10^{-15}	2.92×10^{-10}	8.15×10^{-10}

Both symbolic engines agree with the reference and with each other, and hold the rod to $\sim 10^{-9}$ through repeated dead-centre crossings even at $l/r = 1.05$.

Drake participates differently here, and the reason is a result rather than a choice. Its continuous-mode API accepts the loop-closure constraint and silently ignores it (§5), so a continuous comparison would measure that defect instead of the dynamics. Drake is therefore run in *discrete* mode, where the constraint is honoured — but a discrete plant integrates itself and cannot share the pinned integrator, so it is scored on constraint satisfaction rather than against the same trajectory. The asymmetry is reported rather than smoothed over.

3 Family 3: the cart–pole

3.1 Why a tree

Both earlier families require a constraint or a loop, which is precisely where Drake's continuous API stops being usable. That makes them poor ground for a three-way comparison: one engine is excluded for a reason unrelated to its dynamics. The cart–pole is a *tree* — prismatic cart, revolute pole, no constraint — so every engine participates in its ordinary mode with nothing set aside, and the comparison is genuinely three-way under one integrator.

It remains structurally new: mixed joint types in one mechanism, and a coupled configuration-dependent mass matrix, where the study's two spring systems had constant ones.

3.2 Construction and degeneracy

Cart of mass M at position x; pole of length l carrying point mass m at angle θ from the downward vertical. With $q = (x, \theta)$,

$$M(q) = \begin{bmatrix} M + m & ml\cos\theta \\ ml\cos\theta & ml^2 \end{bmatrix}, \qquad \det M(q) = ml^2\left(M + m\sin^2\theta\right). \tag{3}$$

The determinant is bounded away from zero for a heavy cart, but as the cart is made light *and* the pole passes through vertical it collapses — by 10^6 at $M/m = 10^{-6}$, twice per swing.

This is a different mechanism from the slider–crank's. There the effective inertia collapsed because a transmission ratio vanished. Here the mass matrix itself becomes near-singular because a nearly massless cart cannot resist the pole's horizontal reaction. Both are geometric rather than hand-built, and they degenerate for unrelated reasons, which is why both are worth having.

3.3 Results

Five mass ratios × four initial angles × three engines, 16 probe states each plus a 10 s pinned integration.

M/m	$\det M$ range	**MechanicsDSL**	**SymPy**	**Drake**
10^{-2}	10^2	5.02×10^{-15}	0.00	2.05×10^{-16}
10^{-4}	10^4	7.49×10^{-15}	0.00	5.55×10^{-17}
10^{-6}	10^6	1.25×10^{-14}	0.00	1.96×10^{-16}

Energy drift was $\sim 10^{-10}$ for all three engines at every mass ratio and starting angle. **Drake's residuals are the smallest of the three** — 10^{-16} against 10^{-14} — which is what a recursive rigid-body algorithm should achieve on a two-body tree it was designed for, and is the reverse of the ordering on the pendulum chain, where the direct closed-form solve was tighter.

3.4 Totals

Engine–case pairs across families 2 and 3	72
Disagreements	**0**
Refusals	**0**
Worst residual, any engine, any case	1.3×10^{-14}

4 The one engine-specific defect

4.1 What happens

On a continuous `MultibodyPlant` carrying a distance constraint, Drake 1.56.0 accepts the constraint, finalises, reports `num_constraints() == 1`, and then returns a *diagonal* mass matrix. The rod is absent from the equations; the computed slider acceleration is exactly 0.0 against −0.993 from the closed form. No exception, no warning. In discrete mode the same model is correct, holding the rod to $\sim 10^{-8}$ at $\mathrm{d}t = 10^{-4}$.

4.2 Provenance, from the artefact

2022-12-07	`baeadb5a4`, #18196 introduces `AddDistanceConstraint` together with `if (!is_discrete()) throw`.
2026-02-17	`73c987a60`, #24079 removes that build-time throw.

From `git tag --contains`: guard present **v1.11.0–v1.50.0**, absent **v1.51.0** onward. #24079 merged the day v1.50.0 released and missed that cut — which release dates alone could not have settled.

4.3 The removal was deliberate and the risk was anticipated

#24079's discussion states the reasoning: to allow continuous-time plants to be used with CENIC, the constraint-adding methods cannot throw at build time. The maintainers identified the consequence themselves — with the build-time throw gone, every constraint-sensitive continuous-time method must throw instead, or the plant computes what one maintainer called "wrong answers" (Jeremy Nimmer, #24079, 2026-02-09). The PR added such a throw to `EvalTimeDerivatives` and noted that other constraint-aware methods still need the same treatment, with a manifest opened to work through the public API.

This is therefore an incomplete migration, not a guard dropped carelessly, and the report should say so.

4.4 The precise gap

Method, continuous plant with a constraint	**Guarded?**
`EvalTimeDerivatives`	**throws**
`CalcMassMatrix`	no
`CalcBiasTerm`	no
`CalcGravityGeneralizedForces`	no

The high-level entry point is protected; the three low-level accessors are not. Composing them — `solve`$(M, \tau_g - Cv)$, the natural route when one wants accelerations rather than a simulation — yields unconstrained dynamics silently. This study's own Drake adapter took exactly that route; had it called `EvalTimeDerivatives`, Drake would have caught the misuse on the first day.

4.5 Scope, and what it is worth

Measured in Drake 1.56.0. **Read, not run** on master, whose source shows no continuous-mode rejection either — but #24079's own comment implies continuous support is integrator-dependent, so master's behaviour cannot be settled by reading at all. Any claim about master needs a master build.

Honestly weighed: one open case in a migration the maintainers documented and are working through. It merits an upstream issue and it is the study's only engine-distinguishing silent failure — which is what claim (B) of TR-2026-04 §5.2 required and the single-family study did not have. It is not, by itself, a paper.

5 The exact-number diagnostic

Three times in this study an engine appeared to disagree, and three times the fault was in the comparison. All three shared a signature: **a relative error that was exactly the same suspiciously round number in every case**.

Error	Cause	Cleanly wrong at
4.53 (then exactly 1.00 at dead centre)	Hamiltonian pathway integrates (q, p) and returns $\dot{p}$, compared against $\ddot{q}$	vanished at N=1, where $M = I$
2.5–6.4, growing in N	Drake reports *relative* joint angles; the Lagrangian uses *absolute*	exact at N=1
exactly 2.00, all 20 cases	pole offset rotated about $+y$ mirrors the mechanism; flips the cart's acceleration	—

The last is the clearest. A sign flip gives $|a_d - a_r| = 2|a_r|$, which divides out to exactly 2 wherever $|a_r| > 1$ — so the number is a fingerprint of a mirrored convention, not a measurement of anything physical.

The rule, with three independent confirmations: in a cross-engine comparison, a relative error that is exactly 1.00 or exactly 2.00 across every case is a convention mismatch in the comparison, not a defect in the engine. A second tell is an error that vanishes in a degenerate case ($N = 1$, $M = I$, $\theta = 0$) and grows away from it: genuine derivation errors rarely switch on at a particular system size, whereas convention mismatches characteristically do, because degenerate cases make differing conventions coincide.

Checking this costs two minutes and would have produced three fabricated findings against widely used software had it been skipped.

6 Position of the study, and disclosure

Three problem families, three engines, independent closed-form referees sharing no library with any engine under test. Across derivation, integration, regular and chaotic regimes, amplitude, a thirty-link ladder, constrained pathways, closed loops with dead centres, and mixed joints with collapsing mass matrices: **no engine returned a wrong answer in its ordinary mode of use**, with the single exception of §5.

The engines differ in *reach* — idiomatic SymPy walls at 5 links, MechanicsDSL at 12, Drake handles 30 at constant cost — and in one API guard. They do not differ in honesty.

Artefacts. `reference_slidercrank.py`, `reference_cartpole.py`, `sweep_slidercrank.py`, `sweep_families.p`
`finding_drake_constraint.py` (standalone; imports `numpy` and `pydrake` only), `out/families.json`.
Committed at `f18a514`.

Disclosure. The author is the maintainer of MechanicsDSL, one of the three engines measured. No engine adjudicates itself or any other; the referees share no library with any of them. AI assistance was used throughout. Every figure was produced by executing the committed code.